

# A Positive Answer to Question 4.6 of arXiv:2602.12699

Run fa\_banach\_001, lane 6

June 28, 2026

## Status

This packet gives a candidate full positive answer to Question 4.6 of Basu, Behera, and Seal, arXiv:2602.12699. The question asks whether semi SCS points are stable under binary  $\ell_\infty$ -sums in the direction not covered by Theorem 4.5 of the source paper.

## Source Question

The source paper defines a point  $x_0$  of a bounded set  $C$  to be a *semi Small Combination of Slice point* (semi SCS point) if, for every  $\varepsilon > 0$ , there is a convex combination of slices of  $C$  contained in  $B(x_0, \varepsilon)$ .

The relevant source context is Theorem 4.5 and Question 4.6 on page 8:

**Theorem 4.5.** *Let  $X$  and  $Y$  be Banach spaces. If  $(x, y) \in \text{semi SCS } (B_{X \oplus_\infty Y})$ , then  $x \in \text{semi SCS } (B_X)$  and  $y \in \text{semi SCS } (B_Y)$ .*

*Proof.* Let  $\varepsilon > 0$  and  $(x, y) \in \text{semi SCS } (B_{X \oplus_\infty Y})$ . Then there exists a convex combination of slices, say  $S = \sum_{i=1}^n \lambda_i S_i$  of  $B_{X \oplus_\infty Y}$  such that  $S \subset B((x, y), \varepsilon)$ . Then, for each  $i$ , by Lemma 4.1, we can find a slice  $\tilde{S}_i$  of  $B_X$  and  $y_i \in B_Y$  such that  $\tilde{S}_i \times \{y_i\} \subset S_i$ . Consider  $y_0 = \sum_{i=1}^n \lambda_i y_i$ , then  $\sum_{i=1}^n \lambda_i \tilde{S}_i \times \{y_0\} \subset \sum_{i=1}^n \lambda_i S(B_{X \oplus_\infty Y}, z_i^*, \delta_i)$ . Let  $\hat{x} \in \sum_{i=1}^n \lambda_i S(B_X, x_i^*, \alpha_i)$ . Then

$$(\hat{x}, y_0) \in \sum_{i=1}^n \lambda_i S(B_{X \oplus_\infty Y}, z_i^*, \delta_i) \subset B((x, y), \varepsilon).$$

Thus  $\|\hat{x} - x\| \leq \|(\hat{x}, y_0) - (x, y)\| < \varepsilon$ . So,  $\hat{x} \in B(x, \varepsilon)$ . Thus,  $\sum_{i=1}^n \lambda_i \tilde{S}_i \subset B(x, \varepsilon)$ ; showing that  $x \in \text{semi SCS } (B_X)$ . In a similar way we can show that  $y \in \text{semi SCS } (B_Y)$ .  $\square$

**Question 4.6.** Let  $X$  and  $Y$  be Banach spaces. If  $x \in \text{semi SCS } (B_X)$  and  $y \in \text{semi SCS } (B_Y)$ , does it follow that  $(x, y) \in \text{semi SCS } (B_{X \oplus_\infty Y})$ ?

In text form, Question 4.6 asks:

Let  $X$  and  $Y$  be Banach spaces. If  $x \in \text{semiSCS}(B_X)$  and  $y \in \text{semiSCS}(B_Y)$ , does it follow that  $(x, y) \in \text{semiSCS}(B_{X \oplus_\infty Y})$ ?

## Idea of the Proof

Take convex combinations of slices around  $x$  and around  $y$ . For each pair of component slices, the average functional

$$(u, v) \mapsto \frac{1}{2}f_i(u) + \frac{1}{2}g_j(v)$$

defines a slice of the  $\ell_\infty$ -sum ball. If its aperture is small, then membership in this product slice forces the first coordinate into the  $i$ th slice of  $B_X$  and the second coordinate into the  $j$ th slice of  $B_Y$ . Using product weights  $\lambda_i\mu_j$ , the first coordinate of any resulting convex combination can be regrouped into the original convex combination of slices around  $x$ , and the second coordinate can be regrouped into the one around  $y$ .

## Theorem

**Theorem.** *Let  $X$  and  $Y$  be Banach spaces. If  $x \in \text{semiSCS}(B_X)$  and  $y \in \text{semiSCS}(B_Y)$ , then  $(x, y) \in \text{semiSCS}(B_{X \oplus_\infty Y})$ .*

*Proof.* Let  $\varepsilon > 0$ . Since  $x$  is a semi SCS point of  $B_X$ , there are slices  $S_i = S(B_X, f_i, \alpha_i)$ ,  $1 \leq i \leq n$ , and positive weights  $\lambda_i$  with  $\sum_i \lambda_i = 1$  such that

$$\sum_{i=1}^n \lambda_i S_i \subset B(x, \varepsilon).$$

Similarly, there are slices  $T_j = S(B_Y, g_j, \beta_j)$ ,  $1 \leq j \leq m$ , and positive weights  $\mu_j$  with  $\sum_j \mu_j = 1$  such that

$$\sum_{j=1}^m \mu_j T_j \subset B(y, \varepsilon).$$

We use the standard convention that slices are defined by nonzero functionals, and rescale the functionals so that  $\|f_i\| = \|g_j\| = 1$  for all  $i, j$ .

Put  $Z = X \oplus_\infty Y$ . For every pair  $(i, j)$  choose

$$0 < \eta_{ij} < \frac{1}{2} \min\{\alpha_i, \beta_j\},$$

and define  $h_{ij} \in Z^* = X^* \oplus_1 Y^*$  by

$$h_{ij}(u, v) = \frac{1}{2}f_i(u) + \frac{1}{2}g_j(v).$$

Then  $\|h_{ij}\| = 1$ . Let

$$W_{ij} = S(B_Z, h_{ij}, \eta_{ij})$$

be the corresponding slice of  $B_Z$ . If  $(u, v) \in W_{ij}$ , then

$$\frac{1}{2}f_i(u) + \frac{1}{2}g_j(v) > 1 - \eta_{ij},$$

so

$$f_i(u) + g_j(v) > 2 - 2\eta_{ij}.$$

Because  $u \in B_X$ ,  $v \in B_Y$ , and  $\|f_i\| = \|g_j\| = 1$ , each of  $f_i(u)$  and  $g_j(v)$  is at most 1. Hence

$$f_i(u) > 1 - 2\eta_{ij} > 1 - \alpha_i, \quad g_j(v) > 1 - 2\eta_{ij} > 1 - \beta_j.$$

Thus

$$W_{ij} \subset S_i \times T_j.$$

Consider the convex combination of slices of  $B_Z$

$$W = \sum_{i=1}^n \sum_{j=1}^m \lambda_i \mu_j W_{ij}.$$

Let  $(u, v) \in W$ . Then for some  $(u_{ij}, v_{ij}) \in W_{ij}$ ,

$$(u, v) = \sum_{i=1}^n \sum_{j=1}^m \lambda_i \mu_j (u_{ij}, v_{ij}).$$

For fixed  $i$ , each  $u_{ij}$  belongs to the slice  $S_i$ , and  $S_i$  is convex. Therefore

$$\sum_{j=1}^m \mu_j u_{ij} \in S_i.$$

Consequently,

$$u = \sum_{i=1}^n \lambda_i \left( \sum_{j=1}^m \mu_j u_{ij} \right) \in \sum_{i=1}^n \lambda_i S_i \subset B(x, \varepsilon).$$

The same argument with the roles of  $i$  and  $j$  reversed gives

$$v = \sum_{j=1}^m \mu_j \left( \sum_{i=1}^n \lambda_i v_{ij} \right) \in \sum_{j=1}^m \mu_j T_j \subset B(y, \varepsilon).$$

Hence

$$\|(u, v) - (x, y)\|_\infty = \max\{\|u - x\|, \|v - y\|\} \leq \varepsilon.$$

We have found, for arbitrary  $\varepsilon > 0$ , a convex combination of slices of  $B_{X \oplus_\infty Y}$  contained in  $B((x, y), \varepsilon)$ . Therefore  $(x, y)$  is a semi SCS point of  $B_{X \oplus_\infty Y}$ .  $\square$

## Verification and Novelty Check

The proof is purely formal from the definitions. The key checks are:

1.  $(X \oplus_\infty Y)^* = X^* \oplus_1 Y^*$ , so  $h_{ij} = (f_i, g_j)/2$  has norm one when  $\|f_i\| = \|g_j\| = 1$ .
2. A sufficiently thin slice by  $h_{ij}$  is contained in the product  $S_i \times T_j$ .
3. Slices of a convex set are convex, which justifies the coordinate regrouping.

The run indexes were searched for arXiv:2602.12699, “semi SCS”, and “semi denting”. The only relevant existing row concerns a different counterexample for arXiv:2307.03621 about semi  $w^*$ -denting points and extreme points. Web searches on June 28, 2026, for exact phrases from Question 4.6 and for the source title found only the source arXiv page and no later answer.

## Scope

This answers only Question 4.6, the binary  $\ell_\infty$ -sum converse for semi SCS points. It does not address the source paper's later questions about the case  $1 < p < \infty$  or about converses for projective tensor products.

## References

1. S. Basu, P. P. Behera, and S. Seal, *Semi denting points and related notions in Banach spaces*, arXiv:2602.12699, 2026.